

● MEDIUM

● INFORMATION AND IDEAS

Command of Evidence

06

10 questions · ~15 min

Q1

INFORMATION AND IDEAS

East Australian humpback whales migrate up to 10,000 kilometers each year to reach their breeding grounds. Researchers long believed that migrating whales live only on the extra energy they stored up during the feeding season. But marine biologist Vanessa Pirotta and her team aren't so sure. They analyzed 20 years of observations of the migrating whales made by citizen scientists (members of the public who help collect data for scientific research). The team claims that the whales may not live only on their stored energy during migration.

Which finding, if true, would most directly support the team's claim?

- A. Citizen scientists have observed many different types of marine animals feeding alongside the whales.
- B. Citizen scientists have made many observations of the whales feeding as they migrate to their breeding grounds.
- C. Citizen scientists have made more observations of the whales migrating to their breeding grounds than of the whales returning to their feeding grounds.
- D. Citizen scientists have recently begun to observe the whales migrating to their breeding grounds earlier in the year.

Mean Ratings for Patients after 21 Days

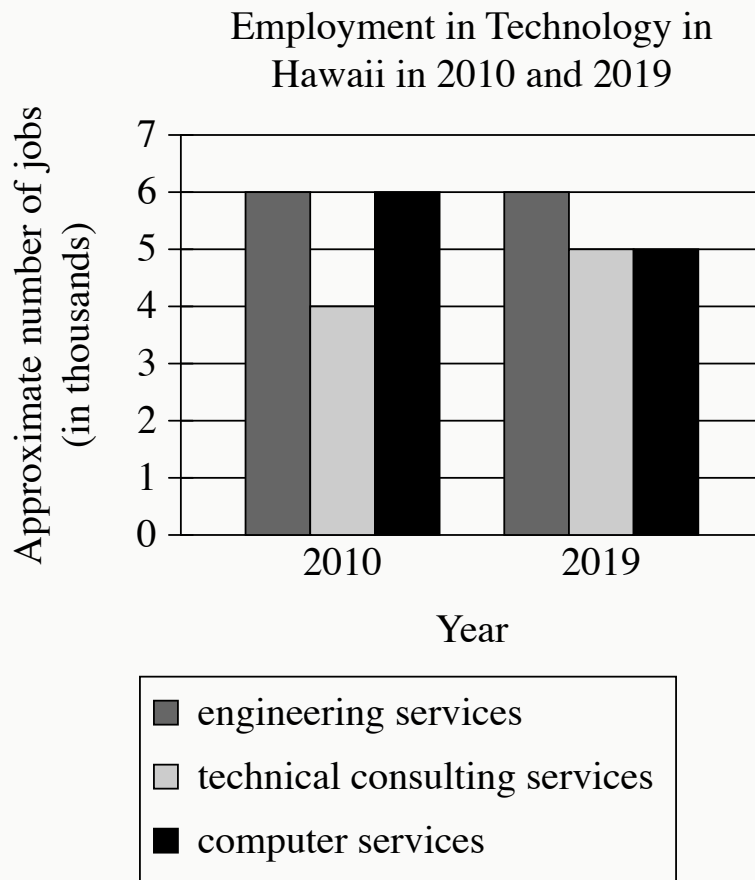
Measure	Mean rating for participants aware of taking a placebo	Mean rating for participants in the control group
Global improvement	5.0	3.9
Symptom severity reduction	92.00	46.00
Quality of life improvement	11.4	5.4

To test whether a medication is effective, scientists compare outcomes for patients taking it and patients taking a placebo (a medically inactive substance). Patients normally aren't told they're receiving a placebo, but a research team conducted a study to see if there might be a medical benefit to telling them. The team used various measures to evaluate participants, with higher ratings indicating greater well-being in each measure. Compared to the mean ratings after 21 days for participants in the control group, the mean ratings for participants who were aware of taking a placebo _____ blank

Which choice most effectively uses data from the table to complete the statement?

- A. ranged from 5.0 to 92.00, indicating that well-being varied widely from participant to participant.
- B. were lower for two measures, with the rating for only one measure indicating greater well-being for these participants.

- C. ranged from 3.9 to 46.00, with no rating indicating greater well-being in any measure for these participants.
- D. were higher for all three measures, indicating greater overall well-being for these participants.



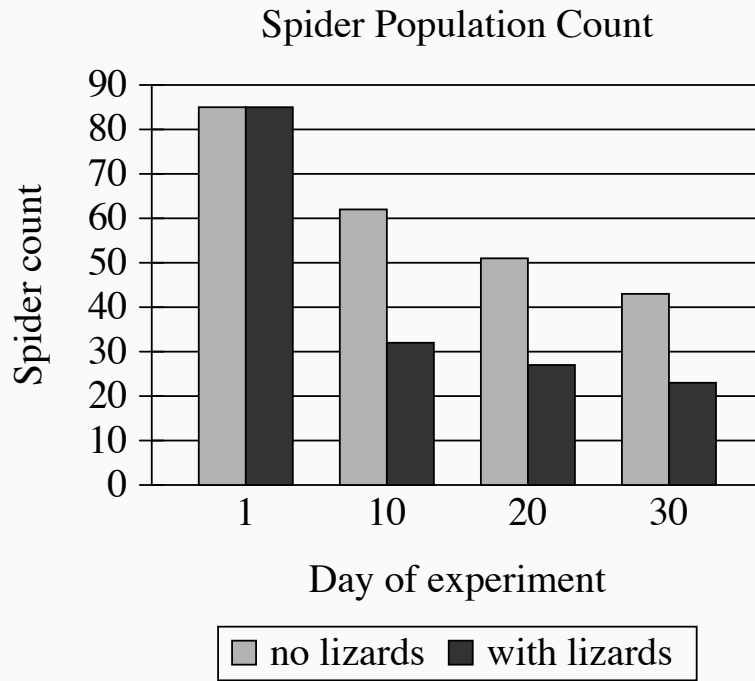
- For each data category, the following bars are shown:
 - engineering services
 - technical consulting services
 - computer services
- All data are approximate.
- The Approximate number of jobs data for the 2 categories are as follows:
 - 2010:
 - engineering services: 6 thousand jobs
 - technical consulting services: 4 thousand jobs
 - computer services: 6 thousand jobs
 - 2019:
 - engineering services: 6 thousand jobs
 - technical consulting services: 5 thousand jobs

- computer services: 5 thousand jobs

A student in Hawaii is interested in pursuing a career in technology and decides to do some research on local trends. The student notices that the number of jobs in computer services in 2010 was _____blank

Which choice most effectively uses data from the graph to complete the statement?

- A. higher than the number of jobs in technical consulting services, and in 2019 was about the same as the number of jobs in engineering services.
- B. about the same as the number of jobs in engineering services, and in 2019 was about the same as the number of jobs in technical consulting services.
- C. lower than the number of jobs in engineering services, but in 2019 was higher than the number of jobs in engineering services.
- D. about the same as the number of jobs in technical consulting services, but in 2019 was lower than the number of jobs in technical consulting services.



- For each data category, the following bars are shown:
 - no lizards
 - with lizards
- All values are approximate.
- The data for the 4 categories are as follows:
 - Day 1:
 - no lizards: 85
 - with lizards: 85
 - Day 10:
 - no lizards: 62
 - with lizards: 32
 - Day 20:
 - no lizards: 51
 - with lizards: 27
 - Day 30:
 - no lizards: 43
 - with lizards: 23

To investigate the effect of lizard predation on spider populations, a student in a biology class placed spiders in two enclosures, one with lizards and one without, and tracked the number of spiders in the enclosures for 30 days. The student concluded that the reduction in the spider population count in the enclosure with lizards by day 30 was entirely attributable to the presence of the lizards.

Which choice best describes data from the graph that weaken the student's conclusion?

- A. The spider population count was the same in both enclosures on day 1.
- B. The spider population count also substantially declined by day 30 in the enclosure without lizards.
- C. The largest decline in spider population count in the enclosure with lizards occurred from day 1 to day 10.
- D. The spider population count on day 30 was lower in the enclosure with lizards than in the enclosure without lizards.

In 1967 the US Congress created the Corporation for Public Broadcasting, which in turn created National Public Radio (NPR). NPR began producing and distributing high-quality news and cultural programming to affiliate stations across the United States in 1971. In a research paper, a student claims that the Corporation for Public Broadcasting and NPR were inspired by the British Broadcasting System (BBC), which had been established in the 1920s.

Which quotation from a work by a historian would be the most effective evidence for the student to include in support of this claim?

- A. “Although the BBC had begun as a private corporation, politicians successfully argued to make it a public company because they believed a public broadcaster could help build national unity in the aftermath of World War I.”
- B. “For many decades, the BBC had no competition since it held Britain’s only broadcasting license, whereas in the United States, the Corporation for Public Broadcasting launched NPR in a broadcasting market already filled with competitors.”
- C. “Congress’s embrace of publicly funded broadcasting reflected a common belief among US politicians that the role of government was not only to ensure people’s safety and liberty but also to enrich people’s lives in other ways.”
- D. “The goal of the BBC was to support British democracy by promoting an informed citizenry, and US legislators believed that ensuring access to high-quality programming could do the same for democracy in the United States.”

Sense and Sensibility is an 1811 novel by Jane Austen. In the novel, Austen describes Marianne Dashwood's ability to persuade others of the rightness of her artistic judgments, as is evident when Marianne visits with John Willoughby, a potential suitor: _____ blank

Which quotation from *Sense and Sensibility* most effectively illustrates the claim?

- A. "Above all, when she heard him declare, that of music and dancing he was passionately fond, she gave him such a look of approbation as secured the largest share of his discourse to herself for the rest of his stay."
- B. "Their taste was strikingly alike. The same books, the same passages were idolized by each—or if any difference appeared, any objection arose, it lasted no longer than till the force of her arguments and the brightness of her eyes could be displayed."
- C. "It was only necessary to mention any favourite amusement to engage her to talk. She could not be silent when such points were introduced, and she had neither shyness nor reserve in their discussion."
- D. "They speedily discovered that their enjoyment of dancing and music was mutual, and that it arose from a general conformity of judgment in all that related to either. Encouraged by this to a further examination of his opinions, she proceeded to question him on the subject of books."

“Odalie” is an 1899 short story by Alice Dunbar-Nelson. In the story, a young woman named Odalie attends the annual Mardi Gras carnival in New Orleans, where she lives with her guardian Tante Louise. Dunbar-Nelson portrays Odalie as eager to escape the monotony of her everyday life: _____ blank

Which quotation from “Odalie” most effectively illustrates the claim?

- A. “Mardi Gras was a tiresome day, after all, she sighed, and Tante Louise agreed with her for once.”
- B. “In the old French house on Royal Street, with its quaint windows and Spanish courtyard green and cool, and made musical by the plashing of the fountain and the trill of caged birds, lived Odalie in convent-like seclusion.”
- C. “When one is shut up in a great French house with a grim sleepy tante and no companions of one’s own age, life becomes a dull thing, and one is ready for any new sensation.”
- D. “It was Mardi Gras day at last, and early through her window Odalie could hear the jingle of folly bells on the [participants’] costumes, the tinkle of music, and the echoing strains of songs.”

Body Length, Filter Time, and Lunges per Dive for Four Whale Species

Whale species	Typical adult body length (meters)	Average time to filter all engulfed water (seconds)	Average number of lunges per dive deeper than 50 meters
fin	18–22	31.30	3.95
humpback	11–17	17.12	6.28
minke	7–10	8.88	7.48
blue	24–34	60.27	4.02

Some whale species practice lunge feeding, in which they lunge toward prey with their mouths open at wide angles, collect the prey and the surrounding water, and then filter out the water through baleen plates in their mouths. Although the volume of water engulfed increases with whales' body length, the surface area of whales' baleen plates, which influences the rate at which water can be filtered, does not increase with body length to the same degree, which helps explain why _____ blank

Which choice most effectively uses data from the table to complete the statement?

- A. minke whales and humpback whales show similar average filter times.
- B. humpback whales show an average of 6.28 lunges per dive.
- C. fin whales show a longer average filter time than minke whales do.

- D. blue whales show the longest average filter time and the highest average number of lunges per dive.

E-book Sales as a Percentage of Total Unit Sales in All Book Formats for a Large US Trade Publisher, by Genre, 2006, 2011, 2016

Genre	2006	2011	2016
science fiction and fantasy	0.6	27.7	36.7
cookbooks	0	2.9	10.5
travel guides	0	5.5	24.6
romance	0.3	40.6	56.2

E-books became an increasingly popular means of reading in the United States in the 2000s and 2010s, though that popularity was concentrated in titles that, like those in most fiction genres, are meant to be read straight through from beginning to end. For books in nonfiction genres that do not tell stories and require the reader to flip back and forth through a volume, e-books were significantly less commercially successful. This can be seen by comparing _____blank

Which choice most effectively uses data from the table to illustrate the claim?

- A. the percentage of 2016 cookbook sales that were e-books with the percentage of 2016 science fiction and fantasy sales that were e-books.
- B. the percentage of 2006 romance sales that were e-books with the percentage of 2016 romance sales that were e-books.
- C. the percentage of 2006 romance sales that were e-books with the 2006 science fiction and fantasy sales that were e-books.

- D. the percentage of 2011 travel guide sales that were e-books with the percentage of 2016 travel guide sales that were e-books.

Dated Ages of Lunar Samples from Select Missions

Mission name	Year	Landing site	Approximate age of lunar samples (billions of years)
Apollo 11	1969	Mare Tranquillitatis	3.6
Apollo 15	1971	Mare Imbrium	3.3
Apollo 17	1972	Mare Serenitatis	3.8
Chang'e 5	2020	Oceanus Procellarum	2.0

The Apollo program missions were spaceflights to the moon led by the United States during the 1960s and 1970s during which astronauts collected some samples of the moon's surface. More recently, China launched the Chang'e 5 mission, which returned additional lunar surface samples. Researchers have analyzed and dated each of the samples, concluding that the lunar samples collected during the Chang'e 5 mission are significant because _____ blank

Which choice most effectively uses data from the table to complete the claim?

- A. they are much younger than the samples brought back from any of the Apollo missions.
- B. they were collected from the same landing site as the Apollo 11 mission.

- C. they are closest in age to the samples brought back by the Apollo 17 mission.
- D. they helped confirm the predicted ages of the lunar samples from the Apollo missions.